

MUSTAFA FATİH CAN

Karsiyaka , Izmir, Turkey | +90 546 934 09 17

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OBJECTIVE

Computer Engineer with 1.5 years of professional experience in backend development and autonomous systems. Competent in building scalable APIs with **Python (FastAPI)** and developing robotics applications using **ROS 2** and **ArduPilot**. Proven experience in **IoT communication (MQTT)** and **geospatial data processing**. Motivated to contribute to defense, IoT, or backend focused software projects with a focus on reliability and performance.

SKILLS AND ABILITIES

- **Languages:** Python, JavaScript, Java, SQL, C++ (Embedded/Arduino).
- **Backend & Robotics:** FastAPI, SQLAlchemy, ROS 2 (Humble), ArduPilot, MAVLink, Spring Boot.
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, RabbitMQ, MQTT, Docker, GCP.
- **Specialized:** GeoServer, GDAL, NumPy, WebRTC, Linux/Bash, REST API, Git.

EXPERIENCE

Computer Engineer - Dronecube

April 2024 – July 2025

- Developed backend APIs using **Python (FastAPI)** following event-driven architecture and **SOLID** principles.
- Implemented **MQTT-based** real-time communication protocols for seamless drone-to-server data exchange.
- Used **Redis** for caching and **RabbitMQ** for asynchronous message processing within distributed services.
- Processed and served geospatial data using **GeoServer** and **GDAL** for agricultural monitoring platforms.
- Managed containerized deployment processes via **Docker** on **Google Cloud Platform (GCP)**.

EDUCATION

Izmir Institute of Technology

2020 – 2025

Computer Engineering Bachelor's Degree

GPA: 3.3 / 4.0

Relevant Coursework: Distributed Systems, Algorithms, Robotics, Operating Systems.

AWARDS AND CERTIFICATIONS

- **Fundamentals of Accelerated Computing with CUDA** – NVIDIA
- **Social Media Content Creator** – Huawei Student Developers

SCHOOL PROJECTS

Bachelor's Thesis: Obstacle Avoidance for Agricultural Drones

- Developed a modular obstacle avoidance system using **ROS 2** and **ArduPilot** for vineyard environments.
- Implemented three distinct algorithms: **Rule-Based**, **Gap-Following**, and **Vector Field Method**.
- Achieved real-time decision-making with a frequency of **10Hz** and latency under **50ms**.
- Conducted performance analysis in simulation environments, generating automated mission reports.

Distributed Database Performance Analysis (Deployed a distributed **MongoDB** cluster across multiple nodes on a local network.)

- Conducted experiments on network delay and data consistency, analyzing the impact of distribution on query performance.

Autonomous Maze-Solving Robot (Designed and programmed a robot using **Arduino Uno** to navigate and solve unknown mazes.)

- Implemented real-time sensor data processing for pathfinding and obstacle detection.

REFERENCES

Selma Tekir

Izmir Institute of Technology

selmatekir@iyte.edu.tr